

HR / POWER

Report di identificazione

Run f78d923aade34b31af005d62edab560a

Dataset: Initial state

Creazione: 2026-10-05T16:19:51.175641+00:00 | Stato: completed

Struttura: p1d_full | $\tau dx/dt + x = K (P(t-L)-P_0)$; $HR_{hat} = B+x$

Inizializzazione: estimated_equilibrium | $P_0=P(0)$; $HR_0=HR(0)$; $x(0)=HR_0-B$; B estimated; $P(t<0)=P_0$

Dataset: 1 segmenti; train richiesti: 1; risultati: 1; validi: 1; falliti: 0.

Ambito identificazione: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

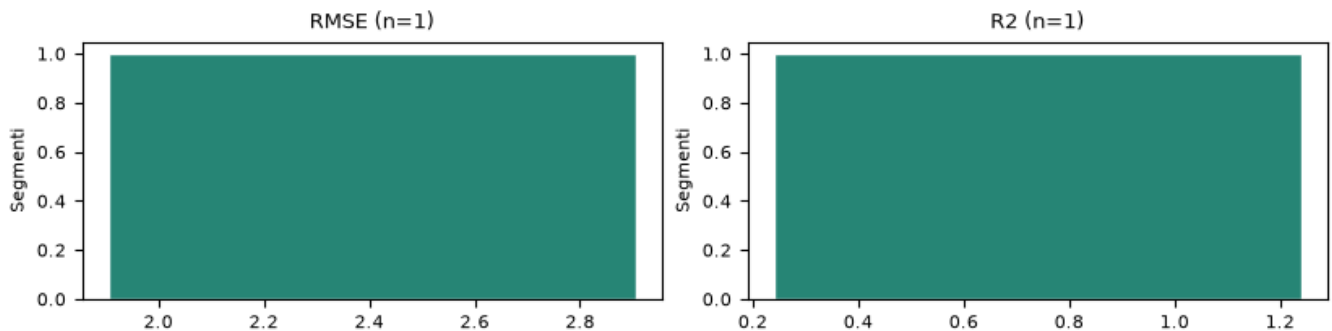
Pre-window richiesta: no; durata: 10 s. P_0 : first_segment_sample; HR_0 : first_segment_sample

Pre-window disponibili: 0 / 1 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R^2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

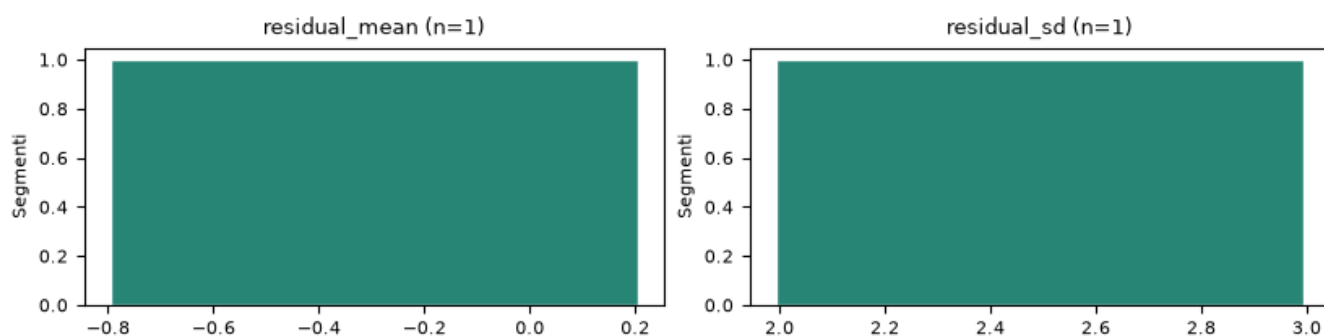
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	1	2.405	2.405	n.d.	2.405	2.405
MAE	1	1.8668	1.8668	n.d.	1.8668	1.8668
bias	1	-0.29592	-0.29592	n.d.	-0.29592	-0.29592
R^2	1	0.73972	0.73972	n.d.	0.73972	0.73972
amplitude_ratio	1	1.13	1.13	n.d.	1.13	1.13



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	1	-0.29592	-0.29592	n.d.	-0.29592	-0.29592
residual_sd	1	2.4929	2.4929	n.d.	2.4929	2.4929
residual_autocorrelation_lag1	1	0.33968	0.33968	n.d.	0.33968	0.33968
residual_autocorrelation_lag5	1	-0.19121	-0.19121	n.d.	-0.19121	-0.19121



Segnalazione	Segmenti
equivalent_predictions_different_parameters	1
multistart_disagreement	1
near_bounds	1
rank_deficient	1

Multistart

Ricerca: profile_multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 10; convergenti: 10. Accordo di tutti gli start: 0 / 1 fit validi.

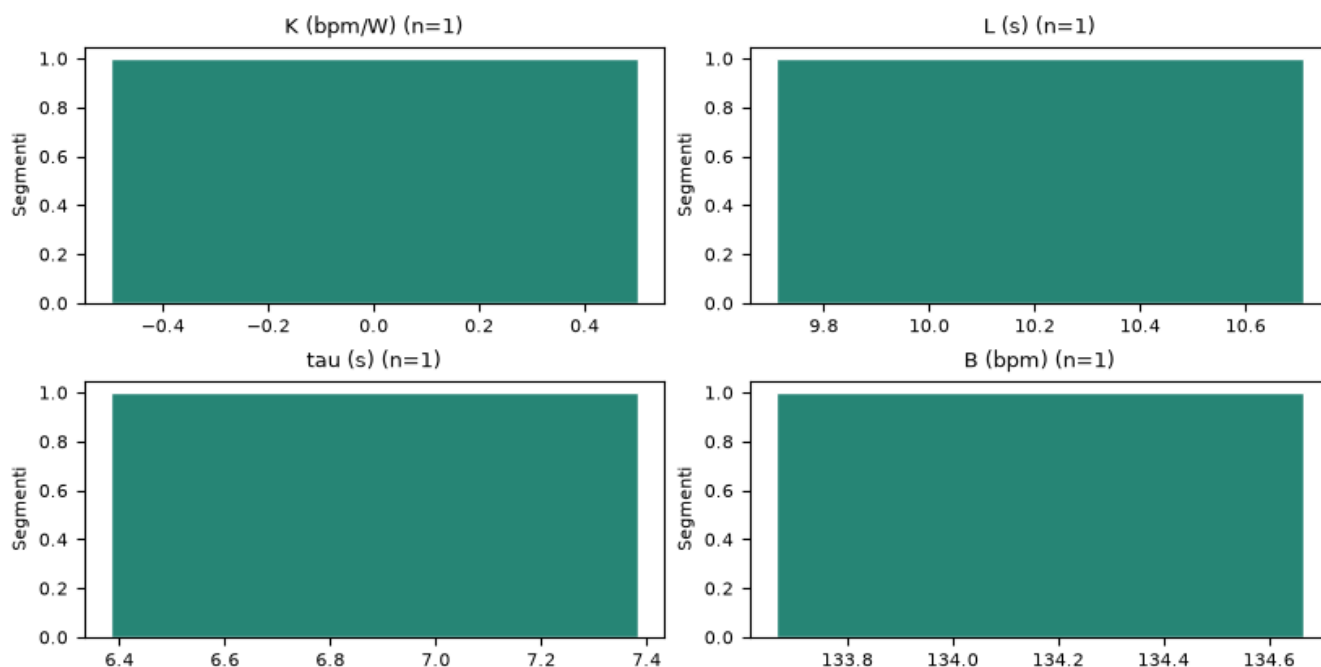
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	10	69.408	69.408	2.0196e-05	69.408	69.408
nfev	10	173.7	179	13.098	138	179

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
K (bpm/W)	1	7.9938e-06	7.9938e-06	n.d.	7.9938e-06	7.9938e-06
L (s)	1	10.209	10.209	n.d.	10.209	10.209
tau (s)	1	6.8825	6.8825	n.d.	6.8825	6.8825
B (bpm)	1	134.16	134.16	n.d.	134.16	134.16

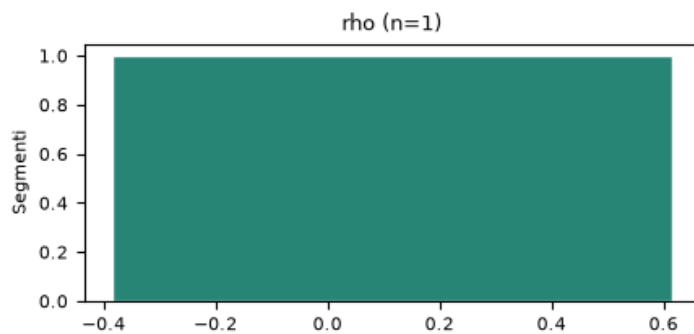
Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
K	n.d.	n.d.	0	1
L	n.d.	n.d.	0	0
tau	n.d.	n.d.	0	0
B	n.d.	n.d.	0	0



Identificabilit 

- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unit  di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.
- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualit  della traiettoria.
- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.
- Multistart: SSE simili con parametri differenti indicano ambiguit . CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	1	0.11498	0.11498	n.d.	0.11498	0.11498
correlation_K_tau	0	n.d.	n.d.	n.d.	n.d.	n.d.
Jacobian rank	1	2	2	n.d.	2	2
Jacobian condition	0	n.d.	n.d.	n.d.	n.d.	n.d.



Scheda segmento

a.csv

ID: 56706faed79a148319eba070 | Stato: fitted

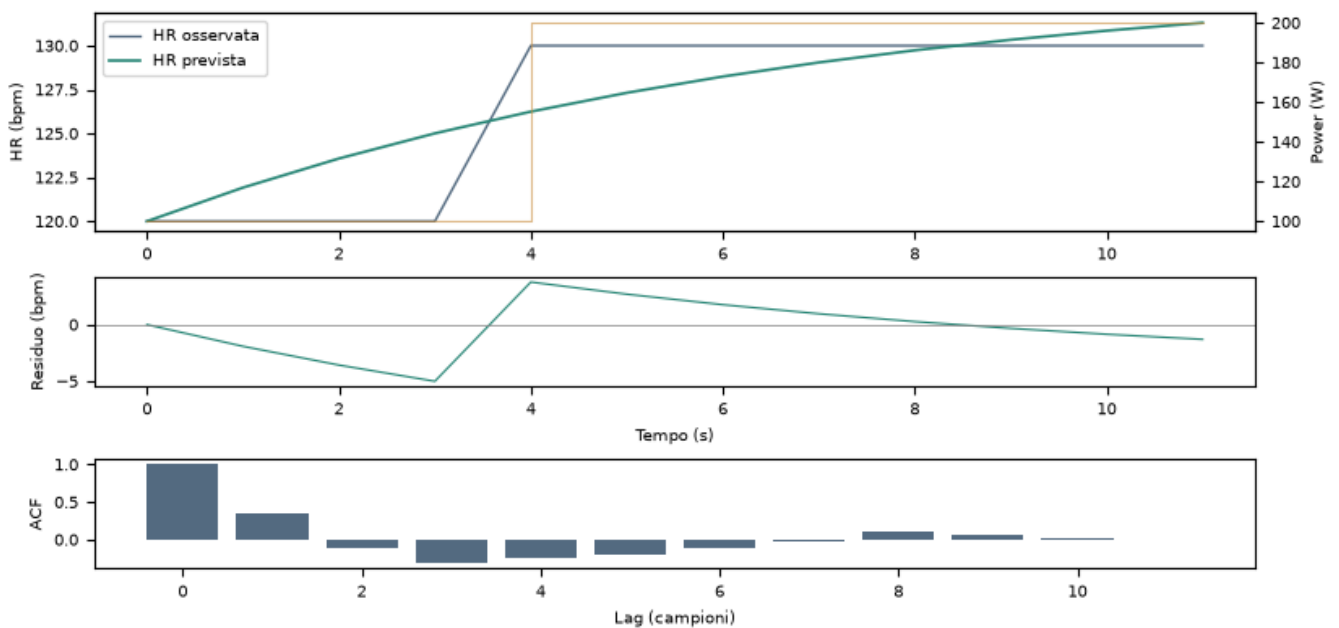
optimizer_success: si | identification_valid: si

Pre-window disponibile: no; usata: no; campioni: 0. P0=100 W; HR0=120 bpm.

HR(0) osservata fissata; B=134.16 bpm stimato; x(0)=-14.164 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.405	1.8668	-0.29592	0.73972	1.13

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.29592	2.4929	0.33968	-0.19121



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	7.9938e-06	n.d.	n.d.	n.d.	no	si
L	10.209	n.d.	n.d.	n.d.	no	no
tau	6.8825	n.d.	n.d.	n.d.	no	no
B	134.16	n.d.	n.d.	n.d.	no	no

rho=0.11498; Jacobian rank=2/4; cond=n.d.; accordo multistart=no.

Correlazioni: {"K_L": null, "K_tau": null, "K_B": null, "L_tau": null, "L_B": null, "tau_B": null}

Multistart: 10 tentativi, 10 convergenti; SSE min/max: 69.408 / 69.408

Griglia: 42441 coppie L/tau; SSE multistart originale: 69.408; selezione: multistart.

Segnalazioni: rank_deficient, near_bounds, equivalent_predictions_different_parameters, multistart_disagreement

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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```

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"code_sha256": "dedd0ace46b46eaf7ccdcf74caa17c33dd86a60f70387dd2191cb49d83c4edce"
}
}

```